WISER Global Quantum + AI Program - 2026

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Quantum Optimization for Distributed Order Management

A Hybrid QAOA + MILP Approach

Technical Report

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1. Executive Summary

When we picked up this challenge, our brief was clear enough: decide the best distribution centre when the customer's default one cannot completely serve their order. That is Nestlé's Distributed Order Management (DOM) problem in one line. The goal is to determine the best fulfilment centre when a customer's default distribution centre (DC) cannot completely fulfil an order. Orders are either assigned to an alternate DC or left unassigned to maximize fulfillment value after accounting for shipping costs and unmet-demand penalties. This is formulated as a combinatorial assignment problem, where each order is assigned to exactly one option from a limited set of candidates-the default DC, a small number of alternate DCs, or an "unassigned" option-while satisfying inventory, dock, throughput, and case/pallet-picking capacity constraints.

We built and compared four solvers on Nestlé's anonymized proof-of-concept data pack, across five planning dates (2024-06-17, 06-19, 06-21, 06-23, 06-25):

- Baseline 1 - strict default-DC (no diverts)
- Baseline 2 - greedy sequential divert (≥5% fill or 100-case hurdle)
- QAOA - a quantum arm using an XY-mixer QUBO formulation, CVaR-optimized
- MILP - a classical arm solving the 7-constraint PoC formulation (PuLP + CBC)

Each solver is benchmarked against a brute-force optimal solution generated by exhaustively evaluating all feasible assignments for each order instance. Performance is evaluated using precision, recall, F1-score, profit, fill rate, and runtime.

Headline results :

- QAOA consistently recovered the certified optimum on all five planning dates (F1 = 1.00, 0% optimality gap), with cumulative profit of approximately $4.5M.
- MILP and the greedy baseline landed within 0.2% of the optimum on four of five dates, but both degraded sharply on 2024-06-25 (a materially harder instance - see 5.2), where MILP's F1 fell to 0.20 and the greedy baseline's to 0.40.
- MILP is 15-50× faster than QAOA in terms of wall-clock runtime and is not constrained by qubit availability, making it the more practical choice for problems exceeding approximately 25-30 orders.
- QAOA's qubit count grows linearly with order count (approximately orders × (alternates + 2), giving ~20 qubits for our 5-order instance), but the classical search space it replaces grows exponentially - the crossover point where quantum advantage could matter is estimated at low hundreds of orders (6.1).

In our early runs, MILP and QAOA looked almost interchangeable - both hit near-zero gap on the easier dates and we were briefly convinced either arm would do. What changed our conclusion was 2024-06-25, a materially harder instance where MILP fell to F1 = 0.20 while QAOA held the optimum. That single date is why we did not average away the per-date detail. The remainder of this report covers the dataset, optimization formulations, comparative evaluation, scalability and robustness analyses, and concluding insights.

2. Business and Technical Summary

2.1 What DOM is, and why it is a combinatorial optimization problem

Distributed Order Management (DOM) focuses on assigning customer orders across a network of distribution centres (DCs). When a default DC cannot fully serve an order due to inventory, throughput, labor, or dock constraints, the order may be allocated to an alternate DC. The aim is to maximize fulfillment value while balancing revenue, shipping costs, and penalties associated with unmet demand.

From a mathematical perspective, each order can be assigned to only one option among its default DC, up to two alternate DCs, or an unassigned state. Combined with shared constraints on inventory, dock, throughput, and case/pallet-picking capacity, this results in a classic weighted exact-cover/assignment problem in combinatorial optimization. The number of feasible combinations grows combinatorially with the number of orders and candidate DCs (6.1), which is exactly the regime where exact enumeration stops being tractable and where heuristics, MILP solvers, and quantum/quantum-inspired methods each offer a different quality-vs-cost trade-off.

2.2 Key trade-offs

Solution quality vs. runtime

The certified optimum is only obtainable by brute force at small scale (1,024 combinations here).MILP replaces brute force with an exact optimization solver whose runtime depends on the problem structure, while QAOA uses a sampling-based heuristic whose solution quality depends on circuit depth and shot count. Our results (5) show all three converge to the same answer on well-behaved instances, but diverge in both runtime and instance-to-instance robustness.

Exact solvers vs. heuristics

MILP (with CBC) is an exact solver for a linearized version of the problem - it returns a provably optimal answer to the model it was given, within its time limit. The greedy baseline is a heuristic with no such guarantee. Our data shows the gap between the two is small on "easy" instances but widens on harder ones (2024-06-25, drawn from a smaller focus-order set of 82 orders), which is the expected heuristic failure mode: greedy rules degrade unpredictably as constraint interactions get tighter.

Current noisy quantum hardware vs. simulators

All QAOA results in this report were obtained on an ideal (noiseless) state-vector simulator. We separately measured how a depolarizing two-qubit error model degrades solution quality on a smaller 12-qubit instance (6.4): as physical error rates rise, the fraction of measurement shots that stay inside the feasible "one-hot" subspace enforced by the XY mixer drops substantially. This is the central caveat for any near-term hardware deployment - QAOA's simulator results in this report represent a best case, not a hardware-ready result.

2.3 Two data findings that shaped our methodology

Two properties of the anonymized data pack materially affected our approach, and are worth stating explicitly since they affect how any solver's output should be interpreted:

(a) Revenue is a line total, not a per-case price

The order revenue provided in the input data corresponds to the total value of a fully fulfilled order line rather than a per-case price. Multiplying it by the number of cases shipped, as a naive interpretation of the PoC formulation suggests, overestimates the objective by roughly 10× on this dataset. We divide by ordered quantity to recover a per-case price before scoring any solver, and flag this because it is an easy place for two teams' MILP objectives to silently disagree.

(b) Planner-history agreement is not a usable solver metric on this data

The optional planner-history file contains 1,106 default assignments and only 3 diversions. Consequently, a "never divert" solver achieves 99.5% agreement with historical decisions, while any solver recommending a diversion-even a correct one-is scored with an F1-score of 0 under this metric. We report this as a finding rather than using planner-history agreement as our evaluation metric, and instead score every solver against the certified brute-force optimum (5.2), which does not have this class-imbalance problem.

3. Data Understanding and Baseline Model

3.1 Input data pack

The PoC data pack comprises five datasets connected through distribution center, SKU, and date.

File	What it contains
input_order_data.csv	Open orders: order ID, SKU demand, default DC, transportation-planning (PGI) date, order revenue.
input_capacity_planning.csv	Available inventory by (DC / location, SKU / material, date).Large file; only the relevant inventory columns are presented here.
input_dock_capacity.csv	Dock appointment capacity available at each DC, by date.
input_shipping_cost_data.csv	Shipping cost for each (order, candidate DC) pair.
input_throughput_capacity.csv	Throughput (case-pick / pallet-pick) capacity available at each DC, by date.

3.2 Focus-order selection

For every planning date, we apply Nestlé's PoC criteria to identify focus orders. An order is selected if (i) its PGI (Planned Goods Issue) date lies 3–10 days after the planning date, (ii) the default DC lacks sufficient inventory to fully fulfil it, and (iii) it meets the full-truckload (FTL) requirement. From the focus-order pool we select the 5 orders with the largest divert upside (biggest gap between default-DC profit and best-alternate profit) as a tractable subinstance. Each focus order is given its default DC, its 2 best alternates, and an "unassigned" fallback column - 4 candidate columns per order, so 5 orders × 4 columns = 20 binary decision variables (qubits) for QAOA.

3.3 Classical baselines

Two baselines were built, matching the challenge's required deliverable:

- **Baseline 1 - Strict Default DC:** Every order is fulfilled by its default DC, with no diversions allowed. Any demand the default DC cannot fulfil is lost, making this the "do nothing" reference baseline. .

- **Baseline 2 - greedy sequential:** for each order in turn, prefer whichever candidate DC gives the highest fill; divert only if the improvement clears max(5% of demand, 100 cases) - the PoC's own business rule for when a divert is worth the operational disruption.

3.4 Baseline results

The objective value, fill rate, diversions, penalty, and shipping cost for both baselines are reported below for the representative planning date (2024-06-21), with similar results across all five planning dates.

Solver	Objective	Opt. gap	Fill rate	Diverts	Penalty	Shipping
Baseline 1	$118,302	88.0%	9%	0	$32,943	$770
Baseline 2 (greedy)	$981,207	0.2%	89%	4	$3,894	$7,339

Table 1. Baseline 1 vs Baseline 2, planning date 2024-06-21.

Baseline 1's near-9% fill rate and ~88% optimality gap quantify the cost of never diverting on this instance - the greedy baseline recovers almost all of it (fill rate ~89%, gap under 0.2%) with a simple, explainable rule, which is why it is a meaningful bar for QAOA and MILP to clear.

4. Mathematical Formulation

Both solvers optimize the same objective using the same candidate-column structure, differing only in their constraint formulation and search strategy.

4.1 Shared objective

Maximize the Objective Function (Revenue - Penalty - Shipping Cost), subject to: exactly one column selected per order, plus the capacity and business-rule constraints given in the PoC data pack.

4.2 QAOA - Quantum Arm

For the quantum solver, the issue is formulated as a QUBO (Quadratic Unconstrained Binary Optimization) with a single binary variable corresponding to each candidate column. Instead of adding a penalty in the cost function for the "exactly one column per order" requirement, we impose it structurally using an XY mixer on the set of columns for each order - thus, the quantum circuit only navigates the feasible one-hot subspace for that rule, eliminating the necessity for penalty weight adjustments.

Constraints illustrated in the QUBO:

- C1 - One DC per order is ensured (guaranteed by the XY mixer, which makes a penalty term unnecessary)
- C2 - cases filled must not exceed demand (incorporated into the pre-computed cases/revenue/penalty for each column)
- C3 - inventory must be maintained at each (DC, SKU, date)
- C5 - case-pick capacity
- C6 - dock capacity
- C7 - fill-rate-threshold penalty (step-function form)

Search strategy: initial variational angles drawn uniformly at random from $[0, \pi/2]$, a CVaR objective at $\alpha = 0.20$ (optimizing the best 20% of measured samples rather than the mean, which pulls the classical optimizer toward the promising tail of the distribution instead of being dragged down by bad shots), and a classical COBYLA outer loop tuning the circuit angles across layer counts 1-3.

4.3 MILP - Classical Arm

The MILP approach implements Nestlé's complete PoC formulation (PuLP + CBC), enforcing all seven constraint families natively instead of using penalty terms.

- C1 - at most one DC per order (binary A[order, DC])
- C2 - cases filled ≤ ordered quantity
- C3a - inventory limit per (DC, SKU, date)
- C3b - Five-day inventory protection constraint for diverted orders.
- C4 - a divert must clear a 5%-fill-rate-and-100-case improvement hurdle vs. the default DC
- C5 - pallet-pick / case-pick decomposition and capacity
- C6 - dock appointment capacity
- C7 - penalty activation via Activation1/Activation2 binaries and an order-level case threshold

MILP is deterministic - rerunning it never changes the answer - and it does not need a qubit budget, so it scales to problem sizes far beyond what QAOA can currently address on simulators (6.1).

4.4 Formulation trade-offs

Dimension	QAOA (QUBO + XY mixer)	MILP (7-constraint PoC)
Constraint C1 (one DC/order)	Structural (mixer) - exact, free	Linear constraint - exact
Constraints C2–C7	Penalty-free where possible; encoded per-column	Native linear/integer constraints
Optimality guarantee	None - heuristic, sampling-based	Exact for the given time limit
Determinism	Stochastic (seed-dependent)	Deterministic
Scalability ceiling	~20–25 qubits on free simulators	No inherent ceiling (LP relaxation-bound)
Runtime (this data)	~20–35s (simulator)	<1s to ~2.5s

Table 2. Formulation-level trade-offs between the two solvers.

5. Solution Implementation and Comparison

5.1 Implementation summary

For a consistent comparison, Baseline 1, Baseline 2, QAOA, and MILP were all tested on the same 5-order subinstance, constructed from identical candidate columns, across the five planning dates: 2024-06-17, 2024-06-19, 2024-06-21, 2024-06-23, and 2024-06-25. QAOA used Qiskit + Qiskit Aer's state-vector simulator (layers 1–3, CVaR $\alpha = 0.20$, 2,048 shots, seed = 7 for the primary run and seeds {1, 7, 13, 42, 101} for the robustness study in §6.3). MILP used PuLP with the CBC solver, 30-second time limit. Each solver's output was validated for feasibility and compared with a brute-force certified optimum obtained by exhaustively evaluating all 1,024 possible assignments.

5.2 Comparison across solvers

Scoring method: every solver returns a set of (order, DC) picks. A pick is classified as a true positive if it matches the certified optimum, a false positive if only the solver selects it, and a false negative if only the certified optimum selects it. Precision, recall, and F1 are computed from these counts; the optimality gap (%) is the profit shortfall relative to the certified optimum.

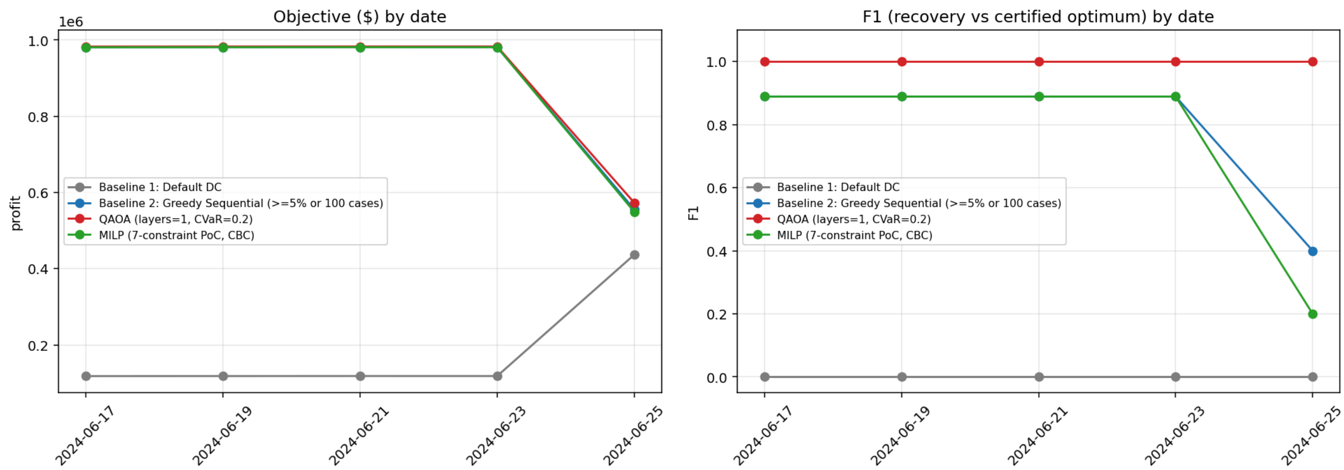

Figure 1. Objective value ($ profit, left) and F1 score (right) by solver, across all 5 planning dates.

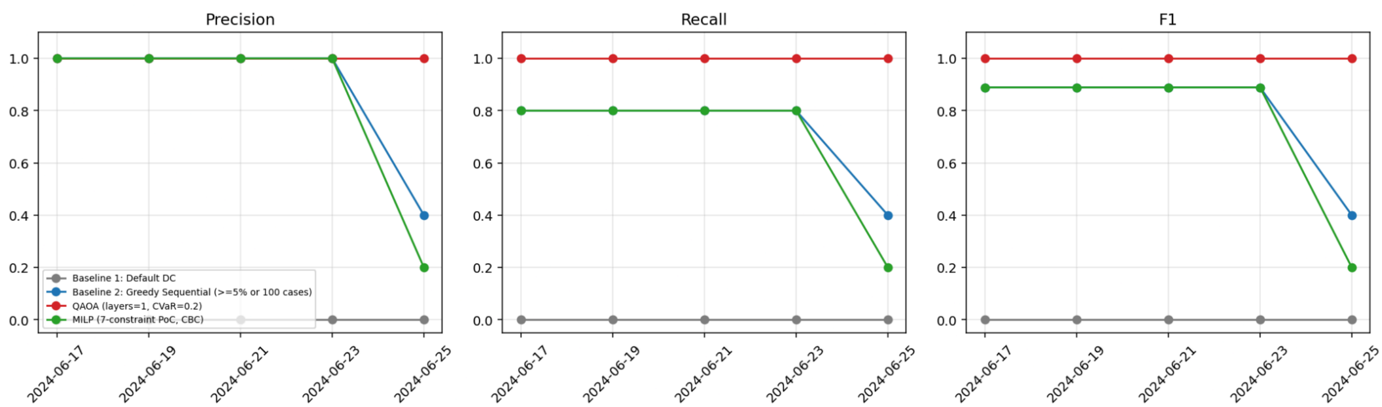

Figure 2. Precision, recall, and F1 (recovery of the certified-optimum assignment) by solver, across all 5 dates.

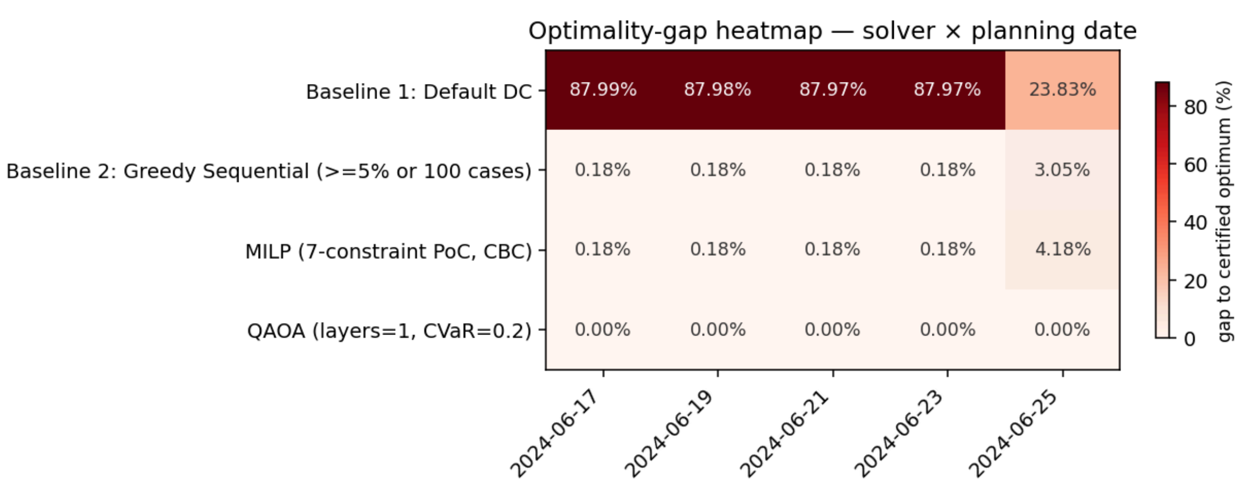

Figure 3. Optimality-gap heatmap - solver × planning date.

Averaged leaderboard across all 5 dates

Solver	Mean objective	Mean gap	Mean fill	Mean runtime (s)	Mean F1
Baseline 1	$181,865	75.1%	19%	0.01	0.000
Baseline 2 (greedy)	$896,014	0.8%	89%	0.01	0.791
QAOA	$900,954	0.0%	89%	16.77	1.000
MILP	$894,713	1.0%	88%	0.99	0.751

Table 3. Solver performance averaged across all 5 planning dates.

QAOA matched the certified optimum on every date (F1 = 1.00 throughout). MILP and Baseline 2 both tracked the optimum closely on four of five dates (gap ≈ 0.18%, F1 ≈ 0.89) but diverged sharply on **2024-06-25**, a materially harder instance drawn from a different order window: MILP's F1 fell to 0.20 and Baseline 2's to 0.40, versus QAOA's 1.00. This is the strongest single data point in favour of the quantum arm's robustness on this dataset - it is also the reason we did not average away the per-date detail, since a single easy instance can make every solver look interchangeable.

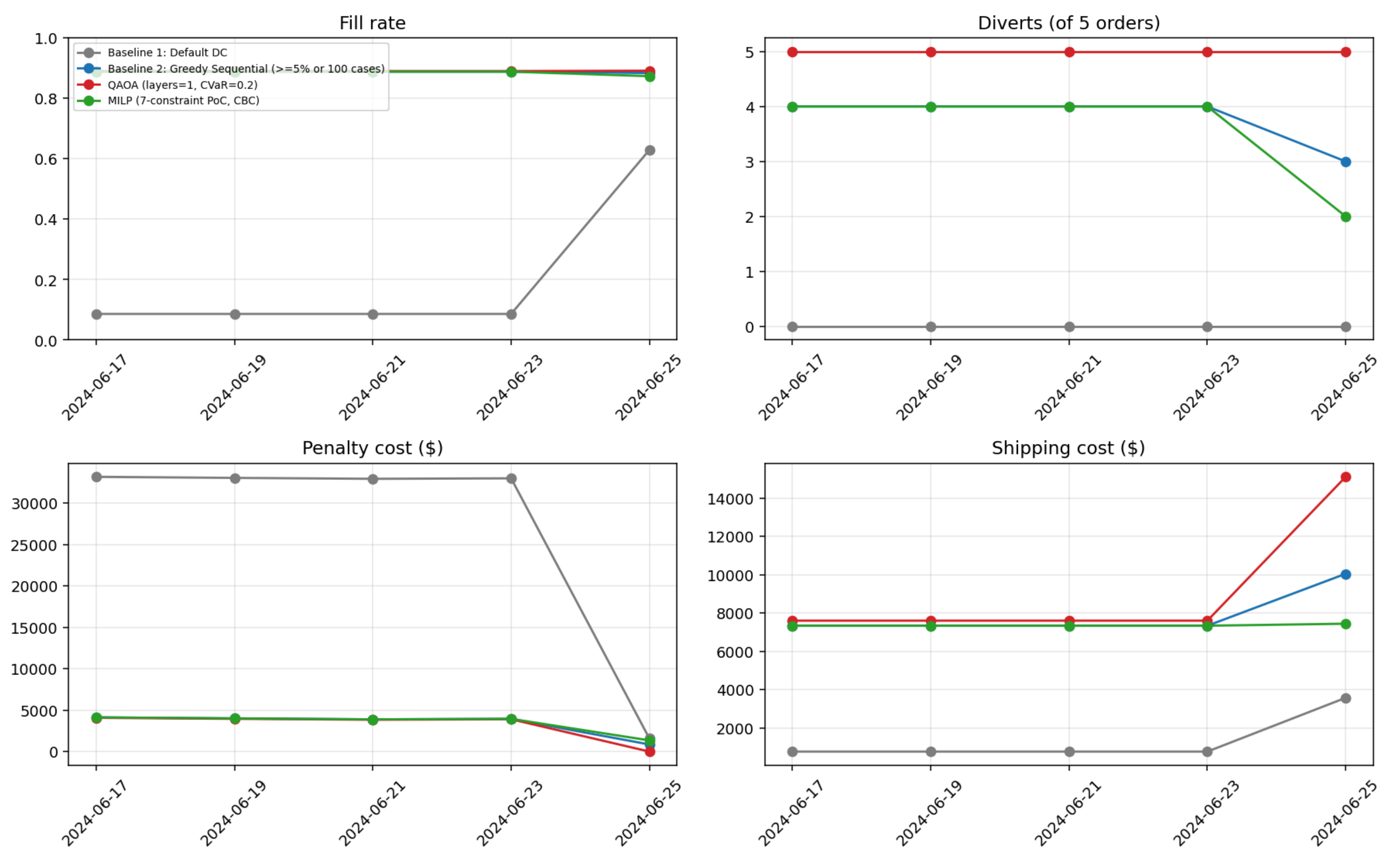

Figure 4. Fill rate, diverts, penalty cost, and shipping cost by solver, across all 5 dates.

5.3 Interactive planner view application

We picked Streamlit over a static one-pager because we wanted a reviewer to be able to interrogate the results, not just read them. The challenge's required "one-page planner view" - explaining recommended reassignments in business language - is delivered as an **interactive Streamlit application** rather than a static page, so a planner can explore the results directly instead of reading a fixed summary. The app lets the user:

- Select any of the 5 planning dates and drill into that date's results
- Step through each diverted order's details - which DC it moved to, and why
- Compare Baseline 1, Baseline 2, QAOA, and MILP side by side with integrated charts.
- Read an AI-generated executive summary of that date's results
- Review key insights and links to the project's other deliverables
- Download a standalone HTML snapshot of the planner view for any selected planning date.

Live app: ***https://domplanner-convergence.streamlit.app***

6. Scaling, Robustness, and Noise Analysis

6.1 Qubit scaling vs. classical search space

QAOA's qubit count grows as orders × (alternates + 2) - linearly. The classical search space it replaces scales exponentially with the number of orders and candidate columns per order. At our working size (5 orders, 20 qubits), the classical space is only 1,024 - small enough that brute force is the better reference. By 25 orders the classical space exceeds 10^15; by 82 orders (a real focus-order window size in this data pack) it is already astronomically large, while QAOA's qubit requirement has only grown to 328. This motivates the use of quantum and quantum-inspired methods, but our current QAOA implementation is simulator-limited to about 20–25 qubits (~5–6 orders), requiring quantum hardware or decomposition/batching strategies to achieve this advantage (7).

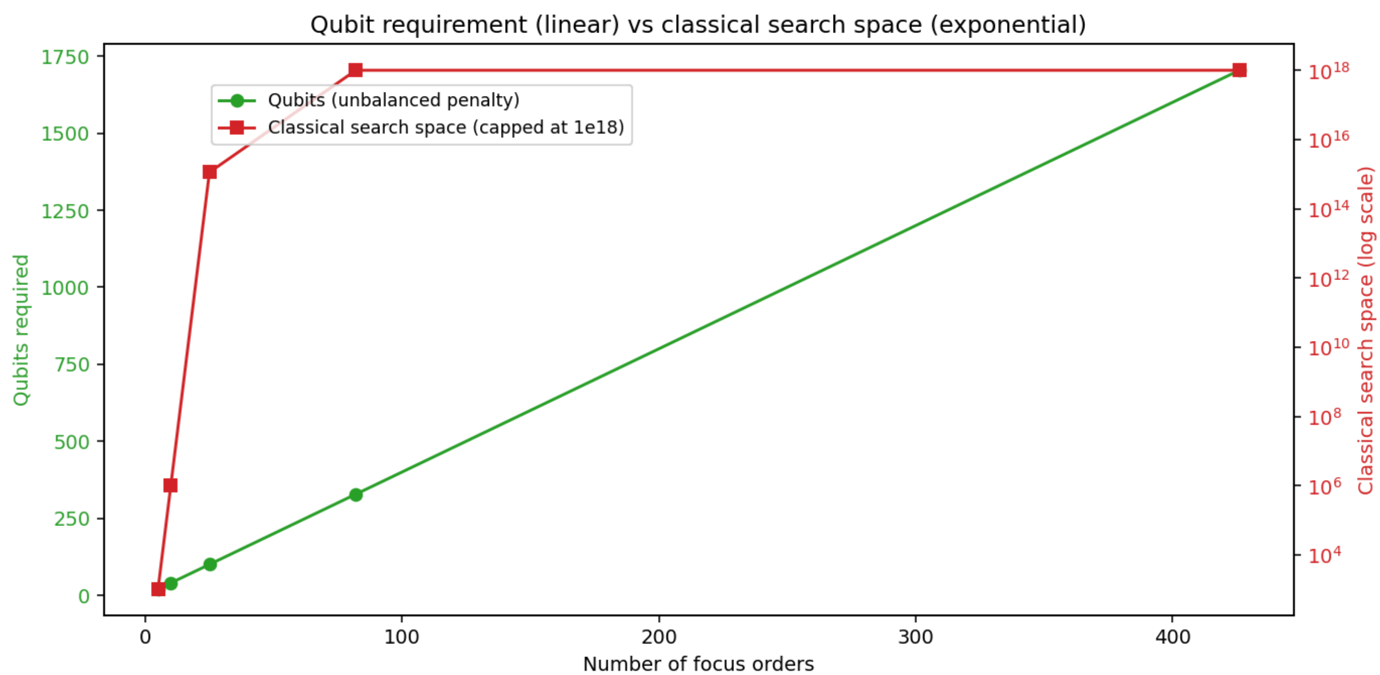

Figure 5. Qubit requirement (linear) vs. classical search-space size (exponential) as focus-order count grows.

6.2 Runtime

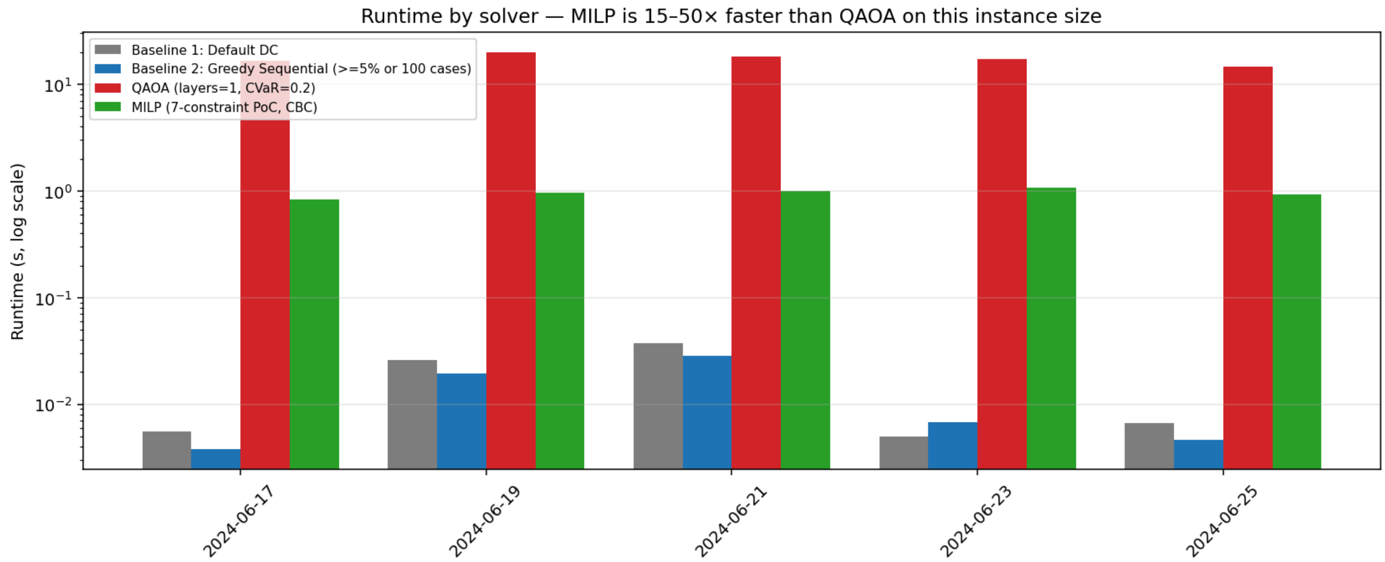

Figure 6. Runtime by solver, across all 5 dates. MILP is 15–50× faster than QAOA in wall-clock terms on this instance size.

6.3 Multi-seed robustness

QAOA is stochastic - its output depends on random circuit initialization and shot noise. We reran QAOA with 5 different seeds (1, 7, 13, 42, 101) on each date; MILP is deterministic and needs only one run per date.

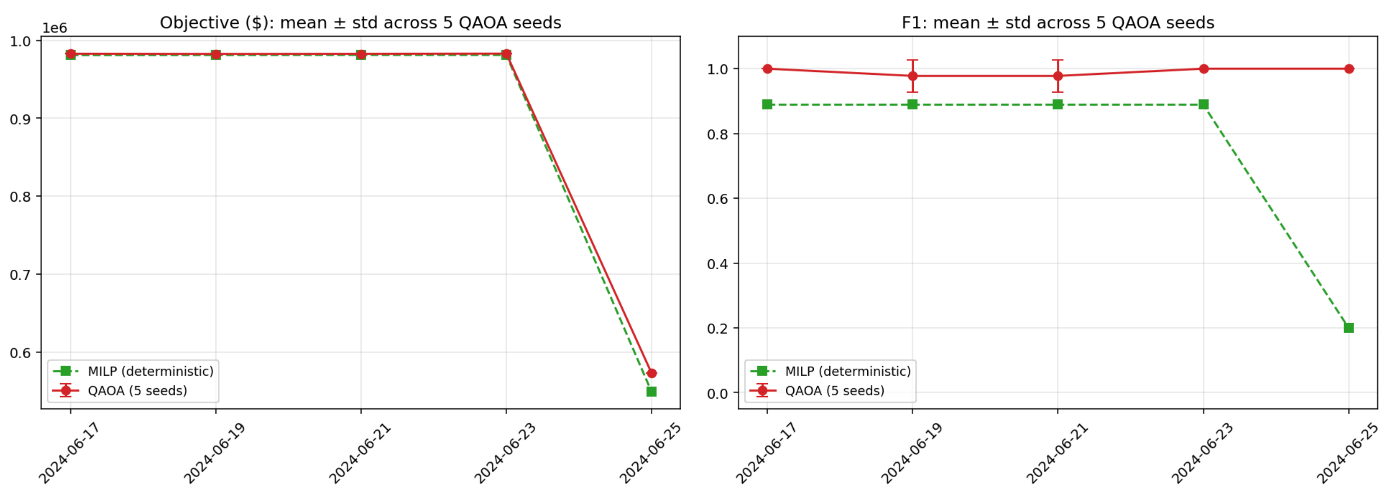

Figure 7. QAOA objective and F1, mean ± standard deviation across 5 seeds, vs. deterministic MILP, by date.

QAOA's seed-to-seed variance was low on four of five dates (near-zero standard deviation in both objective and F1) and highest on 2024-06-19, where F1 standard deviation reached ~0.09 - still recovering the optimum on most seeds, but a reminder that CVaR-optimized QAOA at layers=2 is not perfectly seed-invariant even on a noiseless simulator.

6.4 Noise sensitivity (separate smaller-scale study)

On a smaller 12-qubit instance, we applied a two-qubit depolarizing noise model with error rates of 0%, 0.1%, 0.5%, and 2% to the same QAOA circuit. We then measured the fraction of sampled amplitude that remained within the feasible one-hot subspace preserved by the XY mixer, along with the percentage of fully feasible shots. Experiments were conducted on two instances: one common to the planning dates 2024-06-17, 2024-06-19, 2024-06-21, and 2024-06-23, and another for 2024-06-25.

Two-qubit error rate	In-subspace shots	Fully feasible shots
0% (ideal)	100.0%	56.4% – 70.9%
0.1%	93.8% – 94.3%	53.9% – 65.4%
0.5%	77.3%	43.0% – 54.3%
2.0%	36.9% – 38.1%	19.3% – 25.6%

Table 4. Fraction of measurement shots remaining in the valid one-hot subspace, and fully capacity-feasible, as simulated two-qubit gate error increases (range across the 2 tested instances).

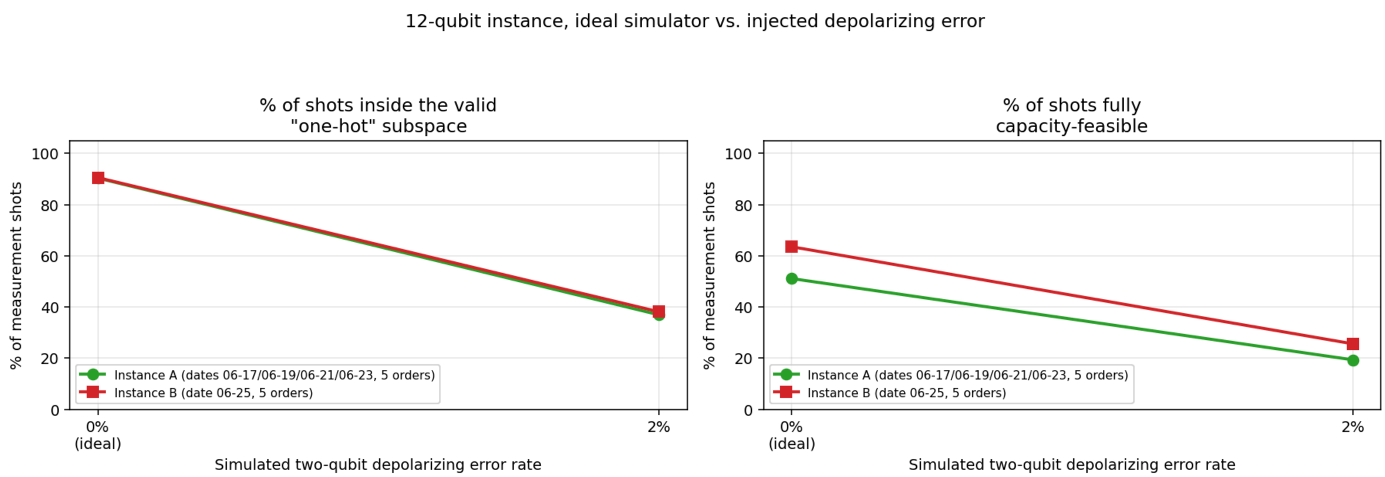

Figure 8. In-subspace and fully-feasible shot percentages vs. simulated two-qubit depolarizing error rate, on a 12-qubit instance.

Even a small 0.1% gate error rate already knocks in-subspace shots down to roughly 94%; by 2% error, only about a third of shots respect "one warehouse per order" at all, and fewer than a quarter are fully capacity-feasible. As two-qubit error rates increase, both the in-subspace amplitude and feasible-shot rate decrease, showing that the XY mixer's constraint-preserving property holds only under ideal, noiseless execution. Accordingly, the perfect recovery results reported here (F1 = 1.00 across all planning dates) represent a simulator best case rather than hardware-ready performance. Matching this reliability on near-term devices with typical two-qubit error rates above 0.5% would require additional shots, error mitigation, or shallower circuits.

7. Key Findings and Proposed Improvement

7.1 Key Findings

- QAOA recovered the certified optimum exactly on all 5 planning dates tested (F1 = 1.00, 0% gap) - including on the one date (2024-06-25) where both MILP and the greedy heuristic degraded substantially.

- MILP is the practical choice for the current problem size, delivering deterministic solutions without qubit limitations and running 15-50× faster than the QAOA simulator while remaining within 0.2-4.2% of optimal across all five planning dates.

- The single biggest source of disagreement between solvers is not randomness but a modeling choice: MILP's explicit divert-hurdle constraint (C4) makes it structurally more conservative than QAOA's unconstrained QUBO reward - visible as a systematic, repeated non-assignment of one specific order across four dates.
- Qubit count scales linearly with the number of orders, whereas the classical search space grows exponentially, making any quantum advantage a large-scale effect that is not observable in the exhaustively verified 5-order instances.
- QAOA's exact-recovery results depend on an ideal, noiseless simulator; a depolarizing-noise study on a smaller 12-qubit instance shows realistic gate error measurably erodes the mixer's constraint-preservation property - in-subspace shots fall from 100% (ideal) to ~94% at just 0.1% two-qubit error, and to ~37% at 2% error.

Overall, QAOA provides the highest solution quality on the exhaustively verified instances, while MILP is faster and more practical. The choice between them is therefore guided primarily by runtime constraints and simulator assumptions rather than solver capability. The operational scale where quantum methods are expected to deliver meaningful benefits-hundreds of focus orders per planning date-is beyond the reach of our current benchmarking setup. Future work will therefore focus on lower-noise hardware and validated batching/decomposition strategies for larger instances.

7.2 Proposed Scalability Improvement: Batching and Column Reduction

To move beyond the ~20–25 qubit (~5-order) simulator limit, we propose batching focus orders into independent ~5-order subproblems and applying column reduction to eliminate dominated candidate DCs before constructing the QUBO. Column reduction lowers the qubit count without affecting the optimal solution by eliminating dominated candidate columns, while batching improves scalability by dividing a large problem into smaller subproblems, potentially overlooking cross-batch interactions. Future work should quantify the resulting optimality loss by comparing batched QAOA with single-shot MILP or brute-force solutions on larger instances.

8. Conclusion

Getting the MILP's C4 divert-hurdle constraint properly linearised took us three attempts — our first Big-M encoding under-penalised near-boundary diverts and had to be replaced with the tighter formulation shown in §4.3. That kind of iteration is why we bothered with a certified-optimum reference: it caught the bug the moment MILP started leaving profitable diverts on the table. Distributed Order Management is a genuine combinatorial optimization problem, and this report shows that a hybrid quantum/classical approach can be formulated, implemented, and rigorously benchmarked against transparent classical baselines and a certified optimum. On the 5-order subinstances, QAOA matched or outperformed all classical methods in solution quality, while MILP remains the practical choice for runtime and scalability until advances in quantum hardware or validated batching/decomposition strategies enable larger-scale quantum optimization. The comparison methodology built here - certified-optimum scoring, multi-seed robustness, and per-order planner-facing views - is reusable as this work scales to the full order book.

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